

THE QUANTUM LANDSCAPE

State of the Art, Architectures
and Installations



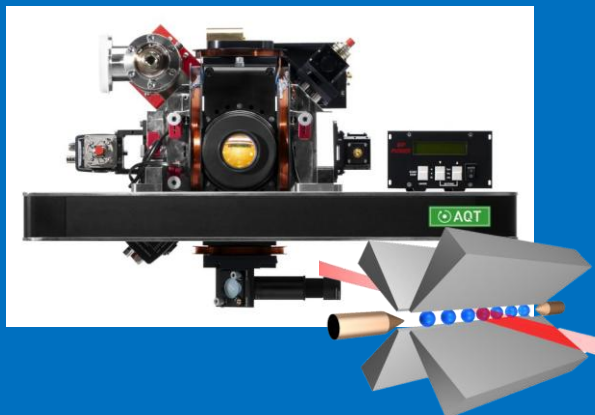
JRC - Ispra
BENQODHI Workshop

10-11 September 2026

The background is a high-angle, wide shot of a city skyline at twilight. The sky is a mix of deep blue and purple, with some clouds catching the last light of the sun. The city below is densely packed with buildings of various heights. Overlaid on the city are several glowing, neon-like lines in red, blue, green, and yellow. These lines are curved and intersecting, suggesting a complex network or data flow. A large, dark blue diagonal shape, resembling a stylized 'V' or a large arrow pointing downwards, is superimposed over the center of the image. The title text is centered within this blue shape.

The Quantum Computing Landscape

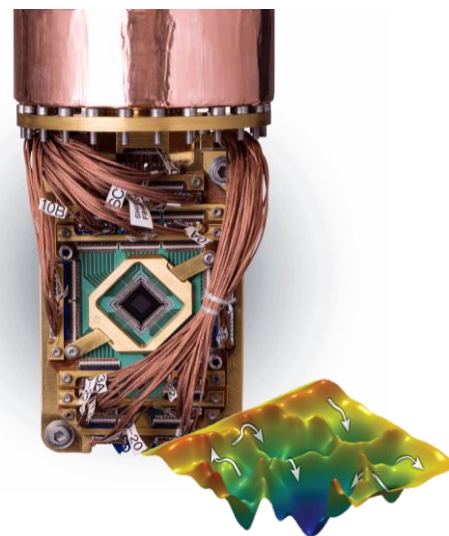
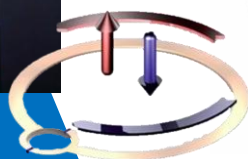
Qubit Types



Trapped Ions qubits

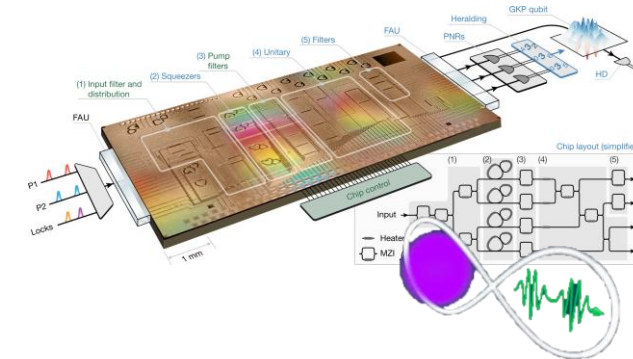


Superconducting qubits

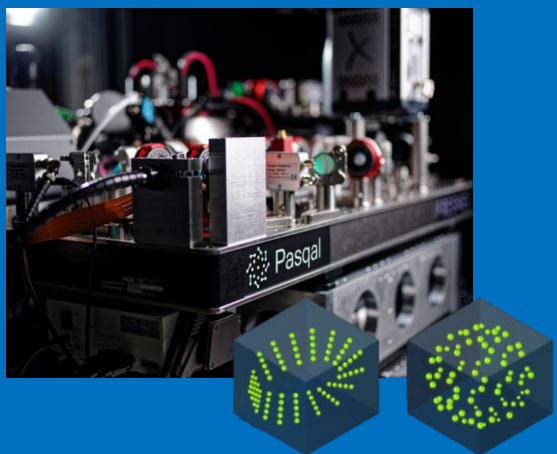


Quantum Annealing

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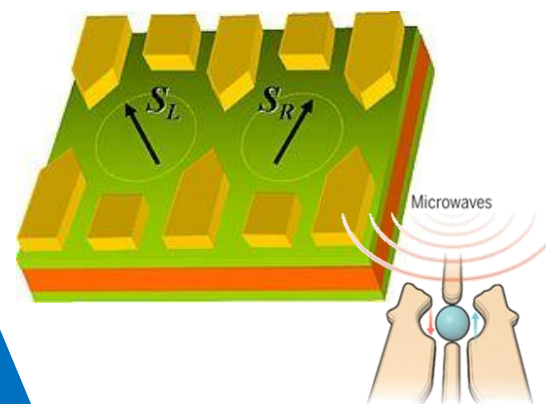
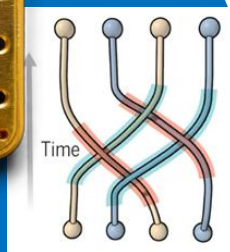
Photon qubits



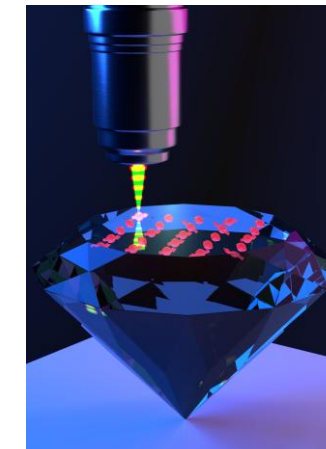
Neutral Atoms qubits



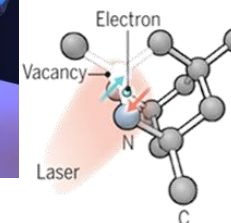
Topological qubits



Quantum dot spin qubits



Nitrogen-Vacancy Center qubits



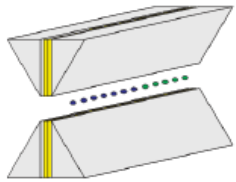
Industry Vendors by Qubit Type

90+ Industry vendors - 8 qubit categories, different subclasses for each category

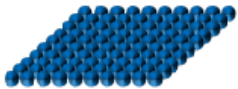
atoms

electrons *controlled spin and microwave cavities*

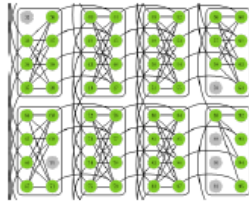
photons



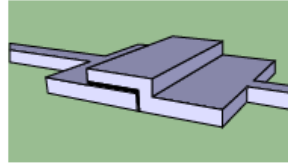
trapped ions



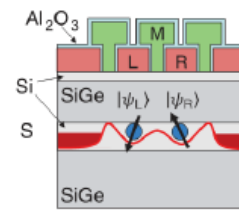
cold atoms



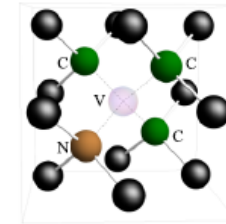
annealing



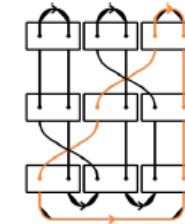
superconducting



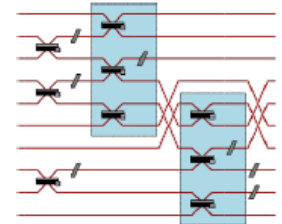
silicon



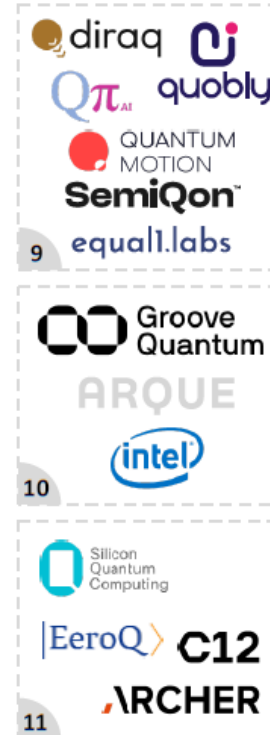
vacancies



topological



photons



Quantum Technology Ecosystem



Quantum Technology Ecosystem

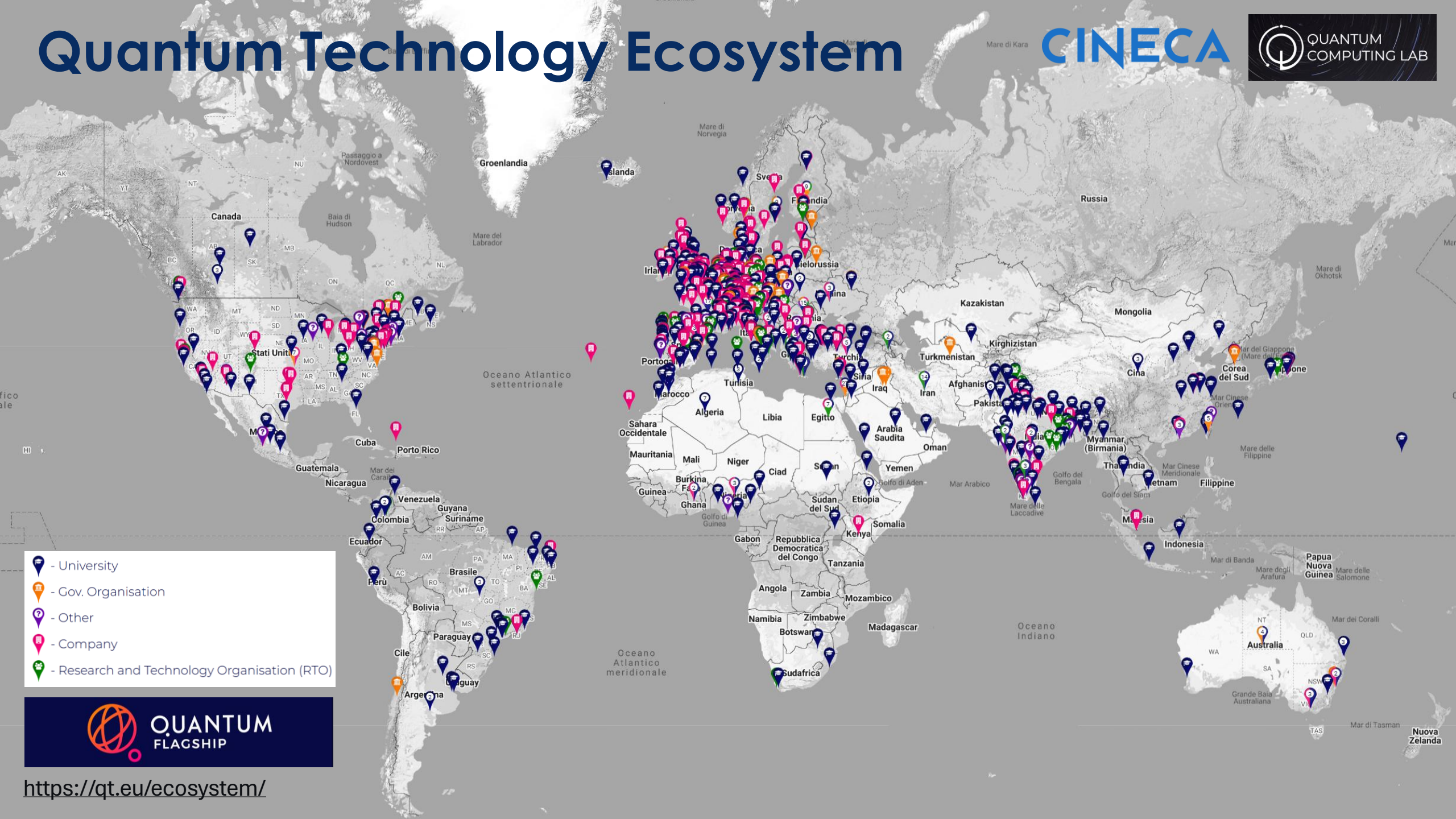
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- University
- Gov. Organisation
- Other
- Company
- Research and Technology Organisation (RTO)



<https://qt.eu/ecosystem/>



EuroQHPC

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LUCY – photonic – 12 qubits



VLQ – superconducting – 24 qubits



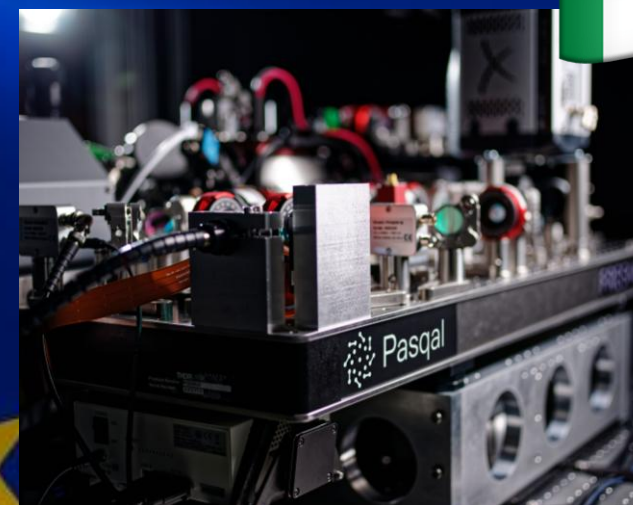
PIAST-Q – trapped-ion – 20 qubits



MareNostrum-Ona – superconducting quantum annealer – 10 qubits



Euro-Q-Exa – superconducting – 54 qubits

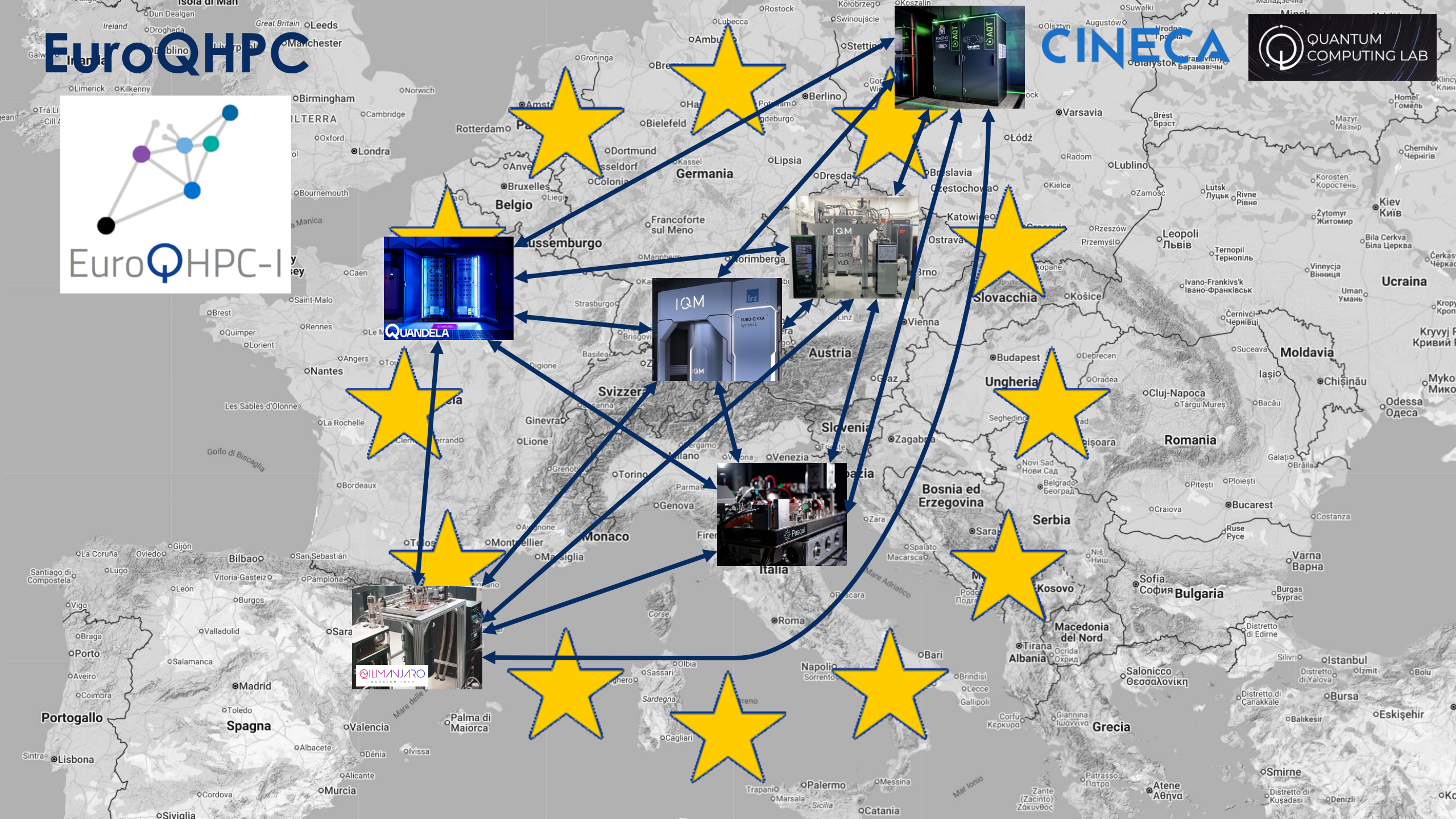


SOL – analog quantum simulator – 140 qubits

EuroQHPC



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EuroQHPC

The European Union is building the world's first public network of cutting-edge quantum computers.

650
qubits
in total
for European
users and science

120
million EUR of EU
& national funds
invested in quantum
research & innovation

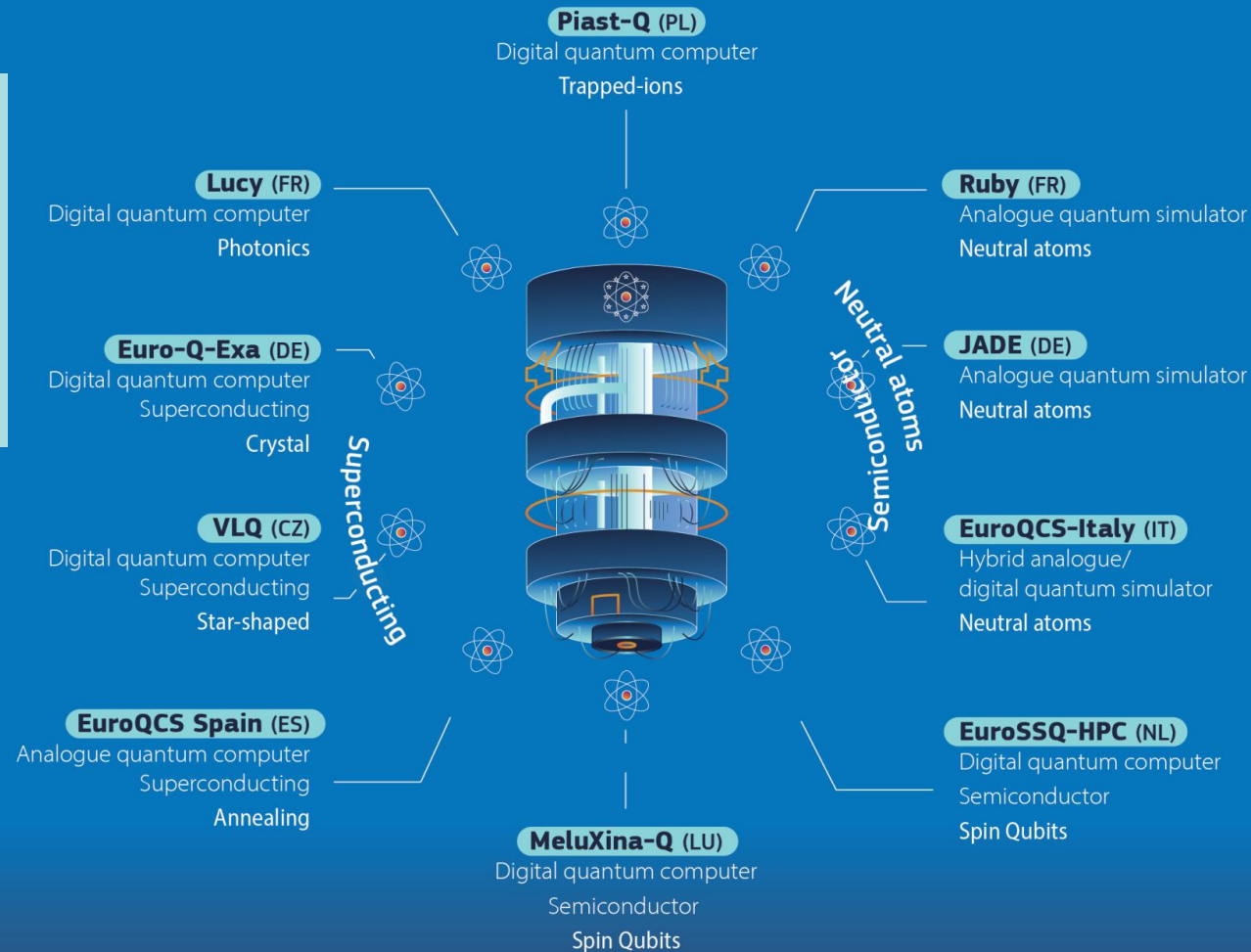
6
different quantum
technologies
are integrated in European
supercomputers

29
partners from 17
European countries
contributed
to the initiative

10
Quantum
Computers



EuroHPC quantum computers
are developed by European companies, and will help
scientists to break unsolvable problems, boosting EU competitiveness,
strategic autonomy and sustainable prosperity.



EuroHPC JU

LEADING THE WAY IN EUROPEAN SUPERCOMPUTING

https://www.eurohpc-ju.europa.eu/quantum-technologies/quantum-computing_en

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Centro Nazionale



CINECA



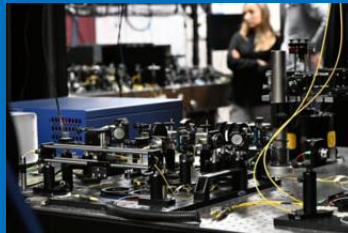
**PoliTo – INRiM –
Links Foundation**



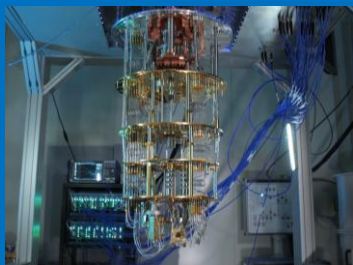
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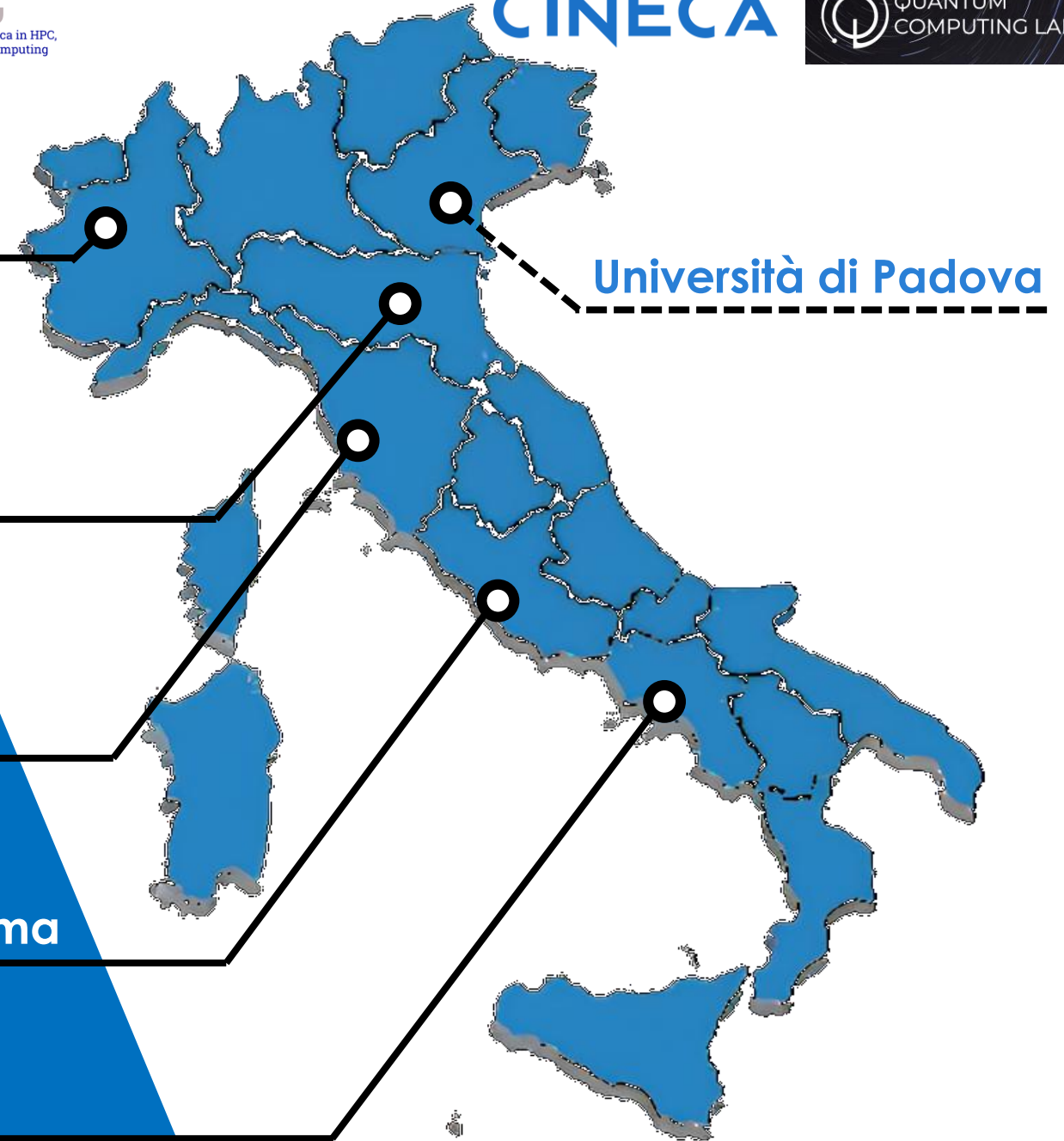
Università di Firenze



**Università
Sapienza di Roma**



**Università degli Studi
di Napoli Federico II**



The background is a high-angle, wide shot of a city skyline at twilight. The sky is a mix of deep blues and purples, with some clouds catching the last light of the sun. Numerous skyscrapers and buildings are visible, their lights beginning to glow. Overlaid on the city are several vibrant, glowing lines in red, orange, yellow, green, and blue. These lines are curved and intersecting, suggesting a complex network or data flow. A large, dark blue diagonal shape cuts across the center of the image, serving as a backdrop for the text.

CINECA Quantum Computers

ONE OF THE MOST POWERFUL HPC FACILITIES IN THE WORLD

Cineca is the most important research supercomputer facility in Italy.



High Performance Computing is a crucial asset for the European Community, to support research and innovation, to **enhance the digital economy.**



ONE OF THE FIRST QUANTUM COMPUTING FACILITIES IN THE WORLD

Cineca is going to host some of the most powerful QCs in Italy.



Quantum Computing is a transformative frontier for the global scientific landscape, to accelerate discovery and breakthroughs, to redefine the limits of computation.

CINECA QCs

European and
National projects



PON QSUD



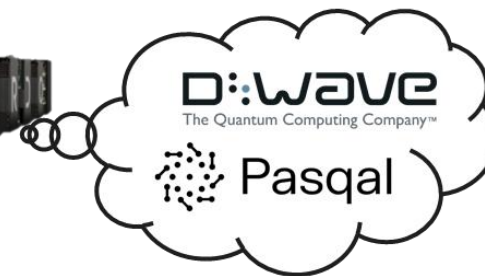
MoSeGad
HPC4UPC



QUANTUM
COMPUTING
AND
SIMULATION
CENTER



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- SOL - Pasqal Orion Gamma 140 qubits
Analog Quantum Simulator  Pasqal
- NOX - IQM Radiance Machine 54 qubits
Digital Quantum Computer  IQM
- Cloud Resources via ISCRA-C
D-Wave Quantum Annealers
Pasqal 100 qubits Quantum Simulator  D-Wave
 Pasqal

NOX - IQM Radiance Machine

- **Superconducting Digital Quantum Computer**
- **54 qubits**
- 90 couplers
- Installed, integration process in progress



Cineca to house Italy's most powerful quantum computer IQM Radiance 54

17/03/2025

🕒 3 min. read

- IQM Radiance 54-qubit full-stack superconducting quantum computer will be integrated into Leonardo, one of the world's fastest supercomputers in Bologna, Italy.
- Cineca intends to use the system for optimisation of quantum applications, quantum cryptography, quantum communication and artificial intelligence quantum algorithms.
- IQM Radiance 54-qubit quantum computer is scheduled for delivery and installation in Q4 2025. The system will be the first and most powerful on-prem quantum computer in Cineca.



SOL - Pasqal Orion Gamma

- Installation in progress
- **Neutral Atom Analog** Quantum Simulator
- **140 qubits**
- Upgrade with **local addressability** next year



CINECA Quantum Computing Lab:

- Research with Universities, Industries and QC startups
- Internship programs, Courses and Conference (HPCQC)



Sara Marzella



Anita Camillini



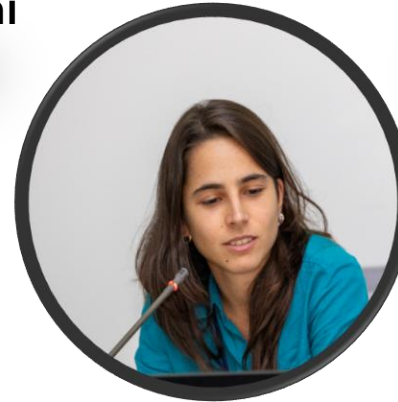
Vito Palmisano



Jonathan
Frassinetti



Gabriele Spada



Francesca Gebbia



Antonio
Costantini

The logo features a large, light blue stylized letter 'C' on a dark blue background. The word 'CINECA' is written in white, bold, uppercase letters across the center of the 'C'.

CINECA

Thank you

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